# LECTURE 9

**Review**

(1) We can rewrite systems of <u>linear eqn's</u> in matrix form $[\,A \mid \vec{b}\,]$.

(2) We learnt how to solve systems $[\,A \mid \vec{b}\,]$ by applying EROs to $[\,A \mid \vec{b}\,]$:

$$[\,A \mid \vec{b}\,] \longrightarrow \dots \longrightarrow \dots \longrightarrow [\,\text{REF} \mid \vdots\,]$$

**Ex**

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 2 & 1 & 1 & -1 \\ 3 & 1 & 1 & 1 \end{array}\right] \longrightarrow \dots \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 1 & 1 \end{array}\right] \begin{array}{l} \Rightarrow x_1 + (-1)x_3 = 2 \\ \Rightarrow x_2 + 3x_3 = -5 \\ \Rightarrow x_3 = 1 \end{array}$$

$$\xrightarrow[R_1 + R_3]{R_2 - 3R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

Thus the sol'n set is $\begin{bmatrix} 3 \\ -8 \\ 1 \end{bmatrix}$.

**Def'n**

An $m \times n$ matrix $A$ is said to be in <u>reduced row echelon form</u> (RREF) if:

(1) it is in REF

(2) the only non-zero entry in a pivot column is the pivot itself

(3) the pivots are all 1s

**Theorem**

Every $m \times n$ matrix $A$ can be row reduced to a <u>unique</u> matrix in RREF.

**Warning**

RREF is unique but REF is not.

**Ex**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ RREF} \qquad \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ not RREF} \qquad \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ RREF}$$

**Ex**

List all possible $2 \times 2$ RREFs.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

reduced row echelon form

# CONTINUED

**Ex**

Solve $\left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 4 \\ 2 & 1 & 4 & 1 & 6 \\ -1 & 4 & -9 & 5 & 4 \end{array}\right]$.

$$\xrightarrow[R_3 + R_1]{R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 4 \\ 0 & 1 & 2 & -3 & -2 \\ 0 & -4 & -8 & -7 & 8 \end{array}\right] \xrightarrow{R_3 + 4R_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 4 \\ 0 & 1 & 2 & -3 & -2 \\ 0 & 0 & 0 & -5 & 0 \end{array}\right] \text{ REF}$$

$$\xrightarrow{R_3(-5)} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 4 \\ 0 & 1 & 2 & -3 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow[R_1 - 2R_3]{R_2 + 3R_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

We get $\begin{cases} x_1 + x_3 = 4 \\ x_2 + 2x_3 = -2 \\ x_4 = 0 \end{cases} \Rightarrow \begin{array}{l} x_1 = 4 - x_3 \\ x_2 = -2 - 2x_3 \\ x_4 = 0 \end{array}$

Hence $x_3$ is a free variable. Let $x_3 = t \in \mathbb{R}$.

Thus the sol'n set is $S = \left\{ \begin{bmatrix} 4-t \\ -2-2t \\ t \\ 0 \end{bmatrix} \;\middle|\; t \in \mathbb{R} \right\}$

$$\left\{ \begin{bmatrix} 4 \\ -2 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}$$

**Def'n**

Let $A \in M_{m \times n}(\mathbb{F})$.
Let $R = \text{rref}(A)$.

(1) The <u>rank</u> of $A$ ($\text{rank}(A)$) is the number of pivots in $R$.

(2) The <u>nullity</u> of $A$ ($\text{nullity}(A)$) is the number of non-pivot cols in $R$.

rank, nullity